

Question 01 – General

(20 Marks – 2 Marks for each)

1. Simplify and find x,

a. $3^{2x} = 9^4$

b. $2^x = \frac{1}{32}$

2. Find the simplified form of the following,

$$\frac{a^2b^{-3}}{a^{-7}b^4}$$

3. Consider the following sets,

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$P = \{1, 3, 7\}$$

$$Q = \{3, 7\}$$

Required:

a) Subsets of Q

b) Proper subsets of P

4. Using a Venn diagram explain the Set Partitioning.

5. Construct the truth table for the following proposition.

$$(\sim p \wedge q) \Rightarrow \sim q$$

6. Construct the truth table to prove the Contrapositive law.

7. Calculate the following,

$$\begin{bmatrix} 5 & 1 \\ -7 & -7 \\ 3 & -8 \end{bmatrix} + \begin{bmatrix} -12 & 11 \\ -3 & -3 \\ 11 & 0 \end{bmatrix}$$

8. Calculate the following,

$$\begin{bmatrix} -2 & 8 \\ 3 & 4 \end{bmatrix} \times \begin{bmatrix} 0 & -7 & -6 \\ 2 & 3 & -5 \end{bmatrix}$$

9. Find the gradient and the interception of the following linear equation.

$$4y + 3x = -28$$

10. Find the center coordinates and the radius of following circle equations.

$$x^2 + y^2 - 4x + 6y + 4 = 0$$

Question 02**(20 marks)**

1. $A = \{x \mid x \in \mathbb{Z}, -2 < x < 7\}$ Retrieve the set elements from the above set builder notation.
(1 mark)
2. In a town of 10000 families it was found that 40% of families buy newspaper A, 20% buy newspaper B and 10% of families buy newspaper C, 5% of families buy A and B, 3% buy B and C and 4% buy A and C. 2% of the families buy all the three newspapers.
(8 marks)
 - a) Number of families buy A only?
 - b) Number of families at least one newspaper?
 - c) Number of families neither buy A nor B?
 - d) Number of families either buy B or C?
3. A relation between two non-empty sets A and B is given as follows $\{(1,5), (2,2), (3,4), (4,3), (5,5), (5,4)\}$
(4 marks)
 - a. Draw the Pictorial Representation of the above relations between Set A (Domain) and Set B (Range).
 - b. Is the above Relation being a Function Justify your answer?
4. Which of the following are partitions of $S = \{a, b, c, d, e, f, g\}$? For each collection of subsets that is not a partition of S, explain your answer.
(3 marks)
 - a. $C_1 = \{\{a, c, e, g\}, \{b, f\}, \{d\}\}$
 - b. $C_2 = \{\{a, b, c, d\}, \{e, f\}\}$
5. If $A = \{a, b\}$, $B = \{1, 2\}$, $C = \{2, 3\}$ find the following
(4 marks)
 - a. $A \times (B \cap C)$
 - b. $(A \times B) \cap (A \times C)$

Question 03**(20 marks)**

1. Write the difference between Primitive Propositions and Composite Propositions using suitable examples. **(2 marks)**
2. Construct truth tables for the followings. **(8 marks)**
 - a. $(\sim p \wedge \sim q) \vee \sim r$
 - b. $(p \Rightarrow q) \wedge (q \Rightarrow p)$
 - c. $[(p \Rightarrow q) \wedge (q \Leftrightarrow p)] \Rightarrow (p \Rightarrow r)$
3. Verify the following using truth tables: $\sim((p \vee q) \wedge r) = (\sim p \wedge \sim q) \vee \sim r$ **(4 marks)**
4. State whether the following compound propositions are tautologies, contradictions, or contingent propositions: **(6 marks)**
 - a. $\sim(p \Rightarrow q) \vee (q \Rightarrow p)$
 - b. $(\sim p \wedge r) \Leftrightarrow \sim(q \vee \sim p)$

Question 04 – Matrices**(20 Marks)**

1. Briefly explain why matrices are important in computing using a suitable example. **(2 Marks)**
2. If $A = \begin{bmatrix} 2 & -4 \\ 3 & -10 \\ -2 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -5 \\ 0 & -2 \\ 7 & -7 \end{bmatrix}$ calculate following **(4 Marks)**
 - a. $A - B$
 - b. $2A + B$
3. If $2A - kB = \begin{bmatrix} -5 & 7 \\ 6 & -14 \\ -25 & 33 \end{bmatrix}$ find the scalar k ? **(4 Marks)**
3. Simplify the following simultaneous equations using matrices. (Note that there will be no marks for any other method instead matrices) **(4 Marks)**

$$5x - y = 19$$

$$4x + 3y = 57$$

4. Find the determinant of the following matrix. **(3 Marks)**

$$\begin{bmatrix} -14 & 2 & 1 \\ 3 & -9 & 1 \\ 15 & 13 & -1 \end{bmatrix}$$

5. There are three types of investigators are available in a private criminal investigation firm in USA. The three types of investigators are Black Hat, White Hat and Grey Hat. Government calls these investigators in different quantities to work on three different cases. Payment for each type of the investigators is different. For the first case 6 Black Hats, 6 White Hats and 2 Grey Hats were used, and total payment is 52 million. For the Second case 6 Black Hats, 3 White Hats and 1 Grey Hats were used, and total payment is 41 million. For the Third case 5 Black Hats, 2 White Hats and 4 Grey Hats were used, and total payment is 39 million. Construct three simultaneous equations for above scenario and solve equations using matrices to find the three different payments for each Black Hat, White Hat and Grey Hat Investigators in millions. **(7 Marks)**

Question 5 – Coordinate Geometry**(20 Marks)**

1. Find the equation of a line(L1) through points (3, 4) and (-4, -6). Now write the equation of the line perpendicular to L1 containing point (2,3). **(2 Marks)**
2. The line M passes through the points A (1, 0) and B (5, 6) and the line N has equation of $2x + 3y = 15$. Lines m and n meet at the point C. **(5 Marks)**
 - a. Find the equation of line M.
 - b. Find the length of line M (hint distance between two points).
 - c. Determine the coordinate of point C.
3. Write the equations of the following circles, **(4 Marks)**
 - a. circle centered at (-4,1) and radius 6.
 - b. circle centered at (2,1) that contains the point (6,4).
4. A military ship locates at point (-4, 7) and ship consist with a weapon having a fire range of 25 meters (assume weapon also on the exact point (-4, 7)). Ship receives ^{four}three enemy targets with their coordinate points A (12, 9), B (14, -12), C (11, -13), D (8, 13) calculate and find which enemy points can be destroyed. **(4 Marks)**
5. Find the equation of the circle that goes through points (4, 3), (-2, -5), (5, 2) and write the center coordinates and the radius. **(5 Marks)**